

1.1: Class Hierarchy Overview

The project uses **Multi-level Inheritance** to organize data efficiently. Inheritance allows us to create a general "template" and then add specific details for different roles without rewriting common code (like name or age).

- **Parent Class (Person):** The base template for any human in the system.
- **Child Classes (Patient & Staff):** These inherit basic info from **Person** but add their own specific details.
- **Grandchild Class (Doctor):** This inherits from **Staff**, meaning it has all the data of a **Staff** member plus specialized medical details.

1.2: Base Class - Person

Every person in the hospital (Admin, Doctor, Staff, or Patient) shares common attributes:

- **Attributes:** Unique ID (**pid**), Name, Age, and Contact Information.
- **Key Method (to_dict):** This converts the Python object into a standard dictionary format. This is critical for saving the data into text files (JSON) later on.

1.3: Child Class - Patient

Inherits from **Person** and adds details necessary for clinical care:

- **Attributes:** Blood group, medical history (a list of past records), and current hospital status.
- **Status Management:** Includes methods like **admit()** (sets status to "ADMITTED") and **discharge()** (sets status to "DISCHARGED").
- **add_history():** Automatically appends new diagnosis and AI treatment notes to the patient's permanent medical record.

This project is a **Smart Hospital Management System** designed to handle daily hospital operations, ranging from patient registration to AI-assisted diagnosis and blood bank management.

It is developed in a structured, modular way across seven major phases. Here is a detailed, phase-by-phase explanation of the project foundation.

Phase 1: Data Models & OOP Architecture

Objective: To define the structural "blueprints" for the hospital system using Python Classes. This phase establishes how data is organized and how different entities (like Patients and Doctors) relate to one another.

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1.4: Staff & Doctor Hierarchy

This sub-phase handles the hospital employees:

- **Staff:** Adds a **role** (like Nurse or Receptionist) and **shift_timing**.
- **Doctor:** A specialized type of staff that includes a **specialization** (e.g., Cardiology) and **available_slots** for booking appointments.

1.5: Standalone Class - BloodInventory

Unlike the "Person" classes, this manages the hospital's blood bank using a powerful library called **NumPy**.

- **NumPy Integration:** Instead of standard lists, blood quantities are stored in a **NumPy array**. This allows the system to perform high-speed calculations, such as instantly filtering for all blood types that fall below a safety threshold (Low Stock).

Phase 2: Backend Persistence (File Handling Engine)

Objective: To implement a robust system that saves all hospital data into JSON files so that information is not lost when the application is restarted.

2.1: File System Structure

The app automatically creates a `data/` folder containing three main files:

- `patients.json`: Stores all patient records and their medical history.
- `staff.json`: Stores the database of doctors and staff members.
- `inventory.json`: Tracks current blood bank quantities.

2.2: The Data Manager Module (`data_manager.py`)

This module serves as the "interface" between the Python code and the hard drive.

- **`load_json`:** Opens the file, reads the data, and turns it into a Python list. It includes **Exception Handling** to ensure that if a file is missing or corrupted, the app doesn't crash but simply returns an empty list.
- **`save_json`:** Receives updated data and writes it back to the JSON files using an "indent" format to keep the files human-readable.

2.3: Serialization & Deserialization

This is the "translation" process for data:

- **Serialization (`to_dict`):** Converts a Python Object (which a computer understands) into a Dictionary (which can be saved as text).
- **Deserialization (`from_dict`):** When the app starts, it reads the JSON text. The `from_dict` method reconstructs those dictionary entries back into fully functional Patient or Staff objects so they can use methods like `admit()` or `add_history()`.

1. Serialization (`to_dict`)

Serialization is the process of converting a complex **Python Object** (stored in your computer's RAM) into a simple **Dictionary** format that can be written as text to a file.

- **Why it is needed:** Computer files cannot "save" a Python class directly; they only understand text.
- **How it works:** In your `models.py`, the `to_dict()` method uses `self.__dict__` to grab all the variables of an object (like Name, Age, and ID) and package them into a standard dictionary.
- **The Result:** This dictionary is then passed to `json.dump()`, which writes it into your `.json` files.

2. Deserialization (`from_dict`)

Deserialization is the reverse process. When you restart your app, it reads the JSON text from your files and turns those dictionaries back into **active Python Objects**.

- **Why it is needed:** When you load data using `json.load()`, Python sees it as a "dead" list of data. You cannot call a method like `.admit()` or `.add_history()` on a plain dictionary.
- **How it works:** You use a `@classmethod` called `from_dict()`. This method takes the raw dictionary data and "reconstructs" a new instance of your class.
- **Restoring Complexity:** It manually restores attributes like `medical_history` or `current_status` so the object is "alive" again and ready to be used by the doctor or staff.

Summary Table for your Viva

Feature	Serialization (<code>to_dict</code>)	Deserialization (<code>from_dict</code>)
Direction	Object → JSON File	JSON File → Object
Goal	Save the data permanently	Load and use the data

Action

Strips away the "methods" and keeps the "data"

Re-attaches the "methods" to the "data"

Syllabus

Link

Mutable Data Structures & File Handling

Advanced OOP (Factory Methods)

The flow of your **Smart Hospital System** follows a structured cycle that moves data from user input to long-term storage, and back into the application. This process ensures that the hospital's "memory" is preserved even after the computer is turned off.

Here is the detailed flow of how data is stored and converted:

1. The Entry Point: Object Creation (RAM)

When a user (like a Receptionist) enters details into a Streamlit form, the application creates a **Python Object** in the computer's temporary memory (RAM).

- **Logic:** The app takes raw strings from the UI and passes them to a class constructor (e.g., `Patient()`) in `models.py`.
- **Validation:** Before the object is fully "born," the code calls `utils.py` to ensure the email is valid and the phone number is exactly 10 digits.
- **Attributes:** The object now exists with its own methods (like `.admit()`) and attributes (like `blood_group`).

2. The Conversion: Serialization (`to_dict`)

Because a hard drive cannot store a "Python Object," we must translate it into a format that a text file understands.

- **The Translation:** The object calls its `to_dict()` method.
- **Data Stripping:** This process strips away the "logic" (the methods) and keeps only the raw "data" (the attributes) in a standard Python Dictionary.
- **Format:** For example, a Patient object becomes: `{"pid": 101, "name": "Derek", "status": "PENDING"}`.

3. The Storage: Persistence (`save_json`)

Now that the data is a simple dictionary, it is handed over to the **Data Manager** to be written to the disk.

- **File Handling:** `data_manager.py` opens the specific file (like `patients.json`) in **Write Mode ('w')**.
- **JSON Dump:** The `json.dump()` function takes the dictionary and converts it into a JSON string.
- **Disk Writing:** The string is physically saved to the `data/` folder on your hard drive.

4. The Retrieval: Deserialization (`from_dict`)

When you restart the application, the flow moves in reverse to bring the hospital's data back to life.

- **Loading:** `data_manager.py` opens the file in **Read Mode ('r')** and uses `json.load()` to pull the list of dictionaries back into memory.
- **The Problem:** At this stage, the data is "dead"—it is just a list of text. You cannot click "Discharge" on a piece of text.
- **Reconstruction:** The app passes each dictionary to a **Factory Method** called `@classmethod from_dict()` in `models.py`.
- **The Result:** This method creates a brand-new object and "fills" it with the saved data. The object is now "alive" again, with all its medical history and hospital methods ready to be used.

Summary Flowchart

1. **UI Input** → Raw strings entered by the user.
2. **Validation** → `utils.py` checks for errors.
3. **Object Creation** → Python Class instance created in RAM.
4. **Serialization** → `.to_dict()` converts Object to Dictionary.
5. **Persistence** → `save_json()` writes Dictionary to `.json` file.
6. **Deserialization** → `from_dict()` reconstructs Object from JSON for the next session.

This code snippet represents the **Deserialization** process within **Phase 2** of your project . Its purpose is to take a raw dictionary (loaded from `patients.json`) and "bring it back to life" as a functional Python Object .

Here is a breakdown of how this code works for your viva:

1. The `@classmethod` Decorator

- **What it is:** This decorator tells Python that this method belongs to the **Class itself** (Patient), not a specific individual patient.
- **Why use it:** It acts as a **Factory Method**. Instead of creating a blank patient and filling in details manually, you call `Patient.from_dict(data)` to manufacture a complete patient object directly from the saved data.

2. Part 1: Initializing the Object

The first block of code calls the standard constructor (`__init__`) to set up the basic identity of the patient .

- **Direct Mapping:** It pulls `pid`, `name`, `age`, and `contact` directly from the dictionary keys .
- **Safety with `.get()`:** For `blood_group`, it uses `.get(..., "Unknown")`. This is a safety feature: if the blood group is missing in the file, it defaults to "Unknown" instead of crashing the program.

3. Part 2: Restoring Complex Attributes

After the basic object is created, the method "rehydrates" the attributes that change over time.

- **`medical_history`:** It restores the list of past diagnoses and AI suggestions. By using `.get(..., [])`, it ensures that a new patient with no history simply gets an empty list.
- **`current_status`:** It restores whether the patient was "ADMITTED" or "DISCHARGED" from the last time the app was used.
- **`assigned_doctor_id`:** It re-links the patient to their specific doctor.

4. Why this is crucial for your project

In your viva, you should explain that without this method, the data loaded from `patients.json` would just be a "dead" dictionary .

- **Dictionary vs. Object:** A dictionary is just a collection of text. It cannot use methods like `.admit()` or `.add_history()`.
- **Functionality:** By using `from_dict`, you convert that text back into an **Object**, which restores all the clinical logic and behaviors defined in your `Patient` class .

Viva Tip: If asked about the difference between this and `to_dict`, you can say: *"to_dict breaks the object down into text to save it (Serialization), while from_dict builds the object back up from text to use it (Deserialization)"*

In this phase, we move beyond basic data entry to add **clinical intelligence** and **workflow automation**. This is the "brain" of the hospital system, where doctors interact with patients and the AI helps with decision-making.

Phase 3: Backend Intelligence & AI Integration

Objective: To assist doctors by using Google Gemini 1.5 Flash to analyze symptoms and suggest treatment plans while maintaining a "Human-in-the-loop" safety protocol.

Sub-phase 3.1: The AI Engine Setup (**ai_engine.py**)

This module configures the connection to the Google AI service.

- **Model Selection:** It specifically uses **Gemini 1.5 Flash**, chosen for its speed and low latency, which is perfect for a real-time hospital application.
- **Error Handling:** The system is built to be robust. If there is no internet or the API key is missing, a **try-except** block catches the error and sets the AI to "Offline Mode" so the app doesn't crash .

Sub-phase 3.2: Prediction Logic & Structured Prompting

This is where the actual "thinking" happens.

- **Structured Prompting:** We don't just ask the AI a general question. We give it a strict persona (Senior Medical Assistant) and provide it with the patient's symptoms and **full medical history** .
- **JSON Enforcement:** To make the data usable in our dashboard, we force the AI to respond **only in JSON format**, including keys for "diagnosis," "suggested_rx" (prescriptions), and "risk_level".

Sub-phase 3.3: The Safety Layer (Human-in-the-Loop)

For ethical and safety reasons, the AI never makes the final decision.

- **Disclaimer Wrapper:** Every AI response is automatically tagged with a warning: *"! AI SUGGESTION ONLY. Doctor must verify before approval"*.
- **Review Process:** The AI output is treated as a **draft**. The doctor must explicitly review, edit, and "Finalize" the notes before they are saved to the patient's permanent file

This code snippet represents **Phase 3: Backend Intelligence & AI Integration** of your project. It defines the **MedicalAI** class, which acts as the "brain" of the application by connecting it to the Google Gemini API to provide diagnostic assistance.

Here is a detailed breakdown for your viva:

1. The Initialization Flow (**__init__**)

The constructor's primary goal is to safely establish a connection with the AI model before the app starts.

- **Security (API Key):** It attempts to retrieve the `GEMINI_API_KEY` from your computer's environment variables. This is a professional practice to keep sensitive keys out of the actual code.
- **Model Configuration:** If the key is found, it uses `genai.configure` and initializes the `gemini-flash-latest` model. This specific model (Gemini Flash) was chosen because it is designed for speed and low latency, making the hospital dashboard feel responsive.

2. The "Safety First" Logic (Exception Handling)

A critical part of this code is the **Try-Except block**.

- **Preventing Crashes:** If something goes wrong—for example, if you have no internet connection or the API key is missing—the `except` block catches the error.
- **The Availability Flag:** Instead of the whole hospital system crashing, it simply sets `self.available = False`.
- **Fallback Readiness:** In your project's larger logic, if `self.available` is false, the UI will automatically switch to a "Manual Mode" where the doctor proceeds without AI help, ensuring the hospital can still function offline .

3. How it Fits into the Workflow

In your viva, you can explain that this class follows the **"Human-in-the-Loop" protocol**:

1. **Request:** The doctor provides symptoms.
2. **Processing:** This class sends those symptoms to Gemini using **Structured Prompting** (which you've defined in the `predict_treatment` method elsewhere) .
3. **Result:** The AI returns a suggested diagnosis and prescription in JSON format, which this class then passes back to the UI for the doctor's final approval .

Viva Tip: If the examiner asks why you chose "Gemini Flash," answer: *"I used Gemini Flash because it provides a balance of medical reasoning and high-speed performance, allowing for real-time diagnostic suggestions without making the user wait".*

This code snippet represents the core of **Phase 3: Backend Intelligence & AI Integration** . It defines how the system uses artificial intelligence to assist doctors with diagnosis and treatment planning.

Objective

The goal of this logic is to use **Google Gemini 1.5 Flash** to analyze a patient's symptoms alongside their past medical history to suggest a clinical path. It follows a **"Human-in-the-Loop"** protocol, meaning the AI only suggests, while the doctor makes the final decision .

Detailed Code Breakdown

1. The Input: Symptoms and History

The function `predict_treatment` takes two main inputs:

- **symptoms**: The current issues the patient is facing (e.g., "High fever and dry cough").
- **patient_history**: The patient's entire past medical record. Passing this allows the AI to consider chronic conditions or past allergies when making a suggestion.

2. Structured Prompt Engineering

The `prompt` variable is the most important part of the intelligence . It uses several advanced techniques:

- **Persona Setting**: It tells the AI to "Act as a Senior Medical Assistant" to ensure the tone and depth of the analysis are professional.
- **Constraint Enforcement**: It strictly demands the output be in **JSON format** with no extra text. This is necessary so the Streamlit frontend can easily read and display the data in different sections (like a table for meds and a card for the diagnosis).

3. API Call and Cleaning Logic

The code sends the prompt to the Google cloud using `self.model.generate_content(prompt)`.

- **Markdown Cleaning**: AI models often wrap JSON code in backticks (e.g., `` `json` ``). This code includes logic to strip those characters so that `json.loads()` can convert the text into a Python dictionary without errors .

4. The Safety Layer (Disclaimer)

Before returning the result to the doctor, the code injects a mandatory **disclaimer** into the data .

- **Logic**: `ai_data["disclaimer"] = "⚠️ AI SUGGESTION ONLY. Doctor must verify before approval."`
- **Purpose**: This ensures that even if the AI is 100% accurate, the system reminds the user that a human professional must validate the advice .

5. Fallback Mechanism (`_fallback_response`)

If the internet is down or the API fails, the `except` block triggers the `_fallback_response()`

- **Purpose:** It returns a "safe" dictionary with values like "Service Unavailable" . This prevents the entire hospital application from crashing just because the AI is offline.
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Viva Tip: Why JSON?

If the examiner asks why you forced the AI to output JSON, your answer should be:

"We use **Structured JSON output** so that the application can programmatically extract specific pieces of information—like risk levels or medicine names—and display them in different parts of the UI, such as charts or text boxes, rather than just showing one big block of unorganized text".

Phase 4: Frontend Integration — Streamlit UI

Objective: To build the interactive, role-based Web Interface that connects users to the backend logic .

Sub-phase 4.1: App Skeleton & Navigation

- **Sidebar Routing:** We use `st.sidebar` to create different sections: **Reception Desk**, **Doctor's Cabin**, and **Blood Bank**.
- **Session State:** We use `st.session_state` to ensure that AI results stay on the screen while the doctor is working. Without this, the results would vanish every time the page refreshed.

Sub-phase 4.2: Doctor's Cabin Interaction

This is the visual implementation of the AI logic from Phase 3.

- **Two-Column Layout:**
 - **Left Column (Assessment):** Where the doctor enters symptoms and clicks "Consult AI" .
 - **Right Column (Treatment Plan):** Where the AI's JSON response is parsed and displayed in a clean card format, including an editable text area for the doctor's final prescription .

Sub-phase 4.3: Blood Bank Visualization

- **Interactive Charts:** We convert the inventory data into a Pandas DataFrame and use `st.bar_chart` to show stock levels visually .
- **Stock Alerts:** If blood levels are too low, the UI instantly displays a large red error banner (e.g., "URGENT ALERT: Low stock for A-") .

This Streamlit application serves as the central command center for a **Smart Hospital Management System**. It integrates data management, role-based security, medical AI, and real-time inventory tracking into a single "wide-layout" web interface.

Here is a breakdown of its core components and logic:

1. Security & Authentication Logic

The application utilizes a "Login Wall" to ensure that only authorized staff can access hospital records.

- **Session State Management:** It uses `st.session_state` to persist the user's login status across browser refreshes .
- **Role-Based Access Control (RBAC):** Once logged in, the system retrieves the user's role (Admin, Doctor, Nurse, or Staff) and dynamically generates the sidebar menu options available to them .

2. Admin Dashboard (Staff Management)

The Admin view provides tools for managing the hospital's human resources:

- **User Creation:** Admins can register new doctors or staff, assigning them specific roles, emails, and passwords .
- **Validation:** It integrates utility functions to ensure emails are correctly formatted and contact numbers are valid before saving them to the database .
- **User Management:** Admins can view a table of all existing users (with masked passwords) and delete accounts as necessary .

3. Reception Desk (Patient Operations)

This module handles the front-end patient lifecycle:

- **Registration:** Staff can register new patients, automatically generating unique IDs and saving their basic medical info (age, contact, and blood group) .
- **Status Triage:** Receptionists can update a patient's status between **PENDING**, **ADMITTED**, or **DISCHARGED** .
- **Appointment Scheduling:** This section allows staff to link patients with specific doctors for a chosen date and time, creating a new `Appointment` record in the system .

4. Doctor's Cabin & AI Integration

This is the clinical heart of the application, designed for physician workflows:

- **Appointment Management:** Doctors see only the appointments specifically scheduled for them .
- **Medical AI Consultation:** A doctor can enter a patient's current symptoms and, with one click, consult a **Medical AI** (using Google Gemini 1.5 Flash). The AI analyzes these symptoms alongside the patient's full medical history to suggest a diagnosis and treatment plan .
- **Finalization:** The doctor reviews the AI's suggestions, makes final edits, and saves the treatment directly into the patient's permanent medical history .

5. Blood Bank & Inventory

The Blood Bank module manages critical life-saving resources:

- **NumPy-Powered Tracking:** The system uses **NumPy arrays** for high-speed tracking of blood units across different blood types .
- **Visual Analytics:** Real-time stock levels are displayed in an interactive bar chart .
- **Automated Alerts:** If any blood type falls below a safe threshold, the system triggers a "Critical Shortage" alert .
- **Emergency Broadcasts:** From the Admin Dashboard, a user can broadcast urgent donation requests via email to all registered patients in the hospital network .

Phase 5: Authentication & Admin Panel

Objective: To secure the application behind a login wall and provide an Admin Dashboard for managing hospital staff with unique credentials .

Sub-phase 5.1: Updating Models for Security (**models.py**)

Before we can log users in, our data blueprints must support security credentials.

- **Attributes:** The **Staff** and **Doctor** classes are updated to include **email** and **password** fields .
- **Password Verification:** A new method, **verify_password(self, input_password)**, is added to the **Staff** class to check if the entered password matches the stored one.
- **Persistence:** The **to_dict()** and **from_dict()** methods are updated to ensure these new security fields are saved to and loaded from **staff.json** .

Sub-phase 5.2: The Authentication Manager (**auth_manager.py**)

This is a dedicated logic module that acts as the "bouncer" for the app.

- **Hardcoded Admin:** A constant `ADMIN_USER` is defined (email: `admin@hospital.com`, password: `admin123`) to ensure the system is never locked out, even if the staff database is empty .
- **Login Function:** This function loads `staff.json`, loops through the users, and returns the user's dictionary if the email and password match. If no match is found, it returns `None` .

Sub-phase 5.3: The Admin Dashboard (`main.py`)

This is a restricted view accessible only to users with the "Admin" role.

- **Staff CRUD:** Admins can create new `Doctor` or `Staff` objects through a form .
- **Uniqueness Check:** Before saving a new user, the system checks `staff.json` to ensure the email is unique, preventing duplicate accounts.
- **User Management:** Provides a table to view all current staff (hiding their passwords for security) and a delete button to remove employees from the system.

Sub-phase 5.4: The Login Wall (Refactoring `main.py`)

The main application script is restructured to enforce security .

- **Conditional Rendering:** The app checks `st.session_state['user']` . If it is empty, only the login form is shown .
- **Session Start:** Once `auth_manager.login()` returns a valid user, that user is stored in the **Session State**, and the app reruns to reveal the sidebar and main content .
- **Role-Based Sidebar:** The navigation menu is filtered based on the user's role. For example, a Doctor only sees the "Doctor's Cabin," while Staff see "Reception" and "Blood Bank" .

Phase 7: The Clinical Loop & Finalization

Objective: To complete the "Doctor's Cabin" workflow, ensuring that AI-assisted diagnoses are properly reviewed by humans and permanently saved to patient records .

7.1: The "My Appointments" Filter

Instead of a doctor seeing every patient in the hospital, this sub-phase introduces specialized filtering to keep the dashboard organized .

- **ID Matching:** The system retrieves the logged-in doctor's unique ID from the `session_state`.
- **Smart Filtering:** It scans `appointments.json` to find only those records where the `doctor_id` matches the current user and the status is still "Scheduled" .

- **Result:** The doctor sees a clean, personalized list of patients they are assigned to treat today .

7.2: Clinical History & AI Context

To make the AI smarter, we improve how data is sent to the Gemini engine.

- **Full History Access:** When a patient is selected, the system displays an expander showing their past medical records (Date, Disease, and past AI predictions) .
- **AI Contextualization:** Instead of just sending current symptoms, the code is updated to pass the **entire medical history** string to the `MedicalAI` class. This allows the AI to spot patterns (like chronic issues) rather than just looking at the symptoms in isolation.

7.3: Closing the Loop (The Save Logic)

This is the most critical technical step: replacing "mock" (fake) logic with real data persistence .

- **Human-in-the-Loop:** The doctor reviews the AI suggestion in an editable text area, modifies the dosage or plan as needed, and clicks "Finalize" .
- **Record Preparation:** A new history record is created containing the current date, the confirmed diagnosis, the final prescription, and the treating doctor's ID .
- **Multi-File Update:** 1. **Patient File:** The new record is appended to the patient's `medical_history` list in `patients.json` . 2. **Appointment File:** The specific appointment is updated from "Scheduled" to **"Completed"** .
- **UI Refresh:** The app calls `st.rerun()`, which automatically removes the treated patient from the doctor's "Upcoming" list .

Phase 8: Advanced Communication (Email Alerts)

Objective: To bridge the gap between hospital management and the outside world through automated notifications .

- **Sub-Phase 8.1: Email Service (`email_service.py`):** Uses Python's `smtplib` to create a service that can send urgent alerts .
- **Sub-Phase 8.2: Low Stock Alerts:** If the Blood Bank (using NumPy logic) detects a shortage, the Admin can click a button to send an alert email to hospital authorities .
- **Sub-Phase 8.3: Donation Broadcast:** Admins can trigger a mass "Donation Request" email to all registered patients, personalized with their names and the specific blood type needed .

Summary for your Viva: You have now moved through the entire lifecycle: **Defining** the data (Phase 1), **Storing** it (Phase 2), **Thinking** with AI (Phase 3), **Visualizing** it (Phase 4), **Securing** it (Phase 5), and **Connecting** all workflows (Phases 6 & 7)

This code represents **Phase 8 (or 12/13 depending on the version): The Automated Communication System**. It allows the hospital to communicate with staff and patients automatically using email.

Here is the detailed explanation for your viva:

1. The Core Technology: **smtplib**

- **What it is:** This is a built-in Python library used to send emails using the **Simple Mail Transfer Protocol (SMTP)**.
 - **Configuration:** The code uses a local server (**localhost**) and port **1025**. In a real viva, you should explain that this is a **Debug/Mock Server** used for testing so that you don't send real emails to real people during your project demonstration .
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2. Function: **send_alert()** (Internal Staff Alert)

This function is triggered when the **Blood Inventory** (using its NumPy logic) detects that a blood group has fallen below the safety limit .

- **Purpose:** It sends a short, urgent warning to the Admin's email.
 - **Message:** It includes specific data points like the **blood group** and the **current quantity** remaining.
 - **Workflow:** Staff see a red error on their screen and click a button to "Send Email Alert to Admin" .
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3. Function: **send_donation_request()** (Mass Patient Broadcast)

This is a more "professional" function used by the **Admin Dashboard** to reach out to the community.

- **Personalization:** It takes the **patient_name** and **missing_blood_type** as arguments. Even though it's a mass broadcast, each email feels personal because it uses the patient's actual name from the **patients.json** file .
 - **The Professional Template:** The body of the email is pre-written with hospital details (Location, Hours) to make it look official .
 - **Broadcast Logic:** When the Admin clicks the "Broadcast" button, the system **loops through every patient** in the database and calls this function for each one .
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4. Safety & Error Handling

- **try...except blocks:** Both functions are wrapped in "Error Handling".
 - **Why it's there:** If the email server is down, the app won't crash. Instead, it prints an error and returns `False`, allowing the Streamlit UI to show a message like "Failed to connect to SMTP Server".
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Viva Questions you might get on this:

- **Q: Why use a port like 1025 instead of a standard one like 587?**
 - **A:** Port 1025 is typically used for local "MailHog" or "Python Debugging Servers." It allows us to test email logic without needing an actual internet connection or a Gmail password .
- **Q: How does the Admin know which patients to email?**
 - **A:** The Admin dashboard loads the `patients.json` file using the **Data Manager** and uses a `for` loop to send an email to every registered patient .

In your project, the connection between a patient and a doctor is handled through the **Appointment Engine**, which acts as a bridge between the two entities in the database.

Here is the technical and logical flow of how they are linked:

1. The Data Structure (`models.py`)

The `Appointment` class is designed specifically to hold "Foreign Keys" (unique identifiers) from both the `Patient` and the `Doctor` .

- **patient_id:** Links to the unique ID of a patient in `patients.json`.
- **doctor_id:** Links to the unique ID of a staff member (with the role 'Doctor') in `staff.json`.
- **Cached Names:** The class also stores `patient_name` and `doctor_name` so the system doesn't have to perform a search every time it wants to display the appointment on the screen.

2. The Linking Process (Reception View)

The connection is physically made in the **Reception Desk** section of `main.py` :

- **Selection:** The staff member uses two dropdown menus (Streamlit selectboxes) .
- **Filtering:** The doctor dropdown is filtered to show only staff members whose role is specifically "Doctor".

- **Object Creation:** Once the "Book Appointment" button is clicked, a new `Appointment` object is created containing both IDs, the selected date, and the time slot .
- **Persistence:** This link is saved permanently into `appointments.json`.

3. The Retrieval (Doctor's View)

When a doctor logs in, the "Doctor's Cabin" uses this link to filter the workload:

- **Querying:** The app retrieves the logged-in doctor's `pid` (ID) from the `session_state`.
- **Matching:** It scans `appointments.json` for any record where the `doctor_id` matches the doctor's current ID.
- **Display:** The doctor is then shown only the patients specifically linked to them, allowing them to start the AI-assisted consultation .

4. Closing the Link

Once the consultation is finished and the doctor clicks "Finalize":

- **History Update:** The treatment is saved to the patient's history .
- **Status Change:** The appointment status in `appointments.json` is updated from "Scheduled" to "**Completed**," which effectively breaks the active link in the doctor's "upcoming" queue.

Would you like me to show you the specific code in `main.py` that filters the appointments for the logged-in doctor?